

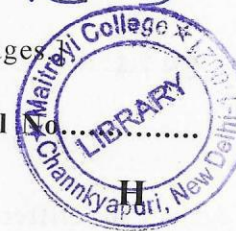
- (b) Explain the modifications that proteins undergo in the lumen of ER. (4)
5. (a) Explain the assembly of microtubules and their role in cellular mobility. Illustrate with the help of suitable diagrams. (8)
- (b) Explain the role of secondary messengers in cell signalling. (4)
6. (a) Discuss the various models of plasma membrane. (6)
- (b) Describe the operation of the Na^+ - K^+ pump with the help of a suitable diagram. (6)
7. Write short notes on any **three** : (4,4,4)
- (a) Synaptonemal complex
- (b) Role of peroxisomes in detoxification
- (c) Functions of Golgi complex
- (d) Receptor-mediated endocytosis

(200)

30.05.24 (M)

[This question paper contains 4 printed pages]

Your Roll No.



Sr. No. of Question Paper : 2974

Unique Paper Code : 32231202

Name of the Paper : Cell Biology

Name of the Course : B.Sc. (H) Zoology (CBCS)

Semester : II

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Answer any **FIVE** questions in all.
3. Question No. 1 is compulsory.

1. (a) Define : (1×4)

(i) Viroid

(ii) Lamin

(iii) Aquaporin

P.T.O.

(iv) Plasmodesmata

(b) Differentiate between the following pairs : (2×5)

(i) Microfilaments and microtubules

(ii) CGN & TGN

(iii) Exocytosis and endocytosis

(iv) Passive and active transport

(v) Mitosis and meiosis

(c) State the contributions of : (1×5)

(i) Gorter and Grendel

(ii) Christian de Duve

(iii) Benda

(iv) Theodore Schwann

(v) Camillo Golgi

(d) Fill in the blanks : (1×5)

(i) _____ chromatin is darkly stained and transcriptionally inactive.

(ii) _____ is known as programmed cell death.

(iii) _____ is the marker enzyme of endoplasmic reticulum.

(iv) Creutzfeldt -Jakob disease (CJD) is caused by _____

(v) _____ is also known as the resting phase of cell cycle.

(f) Expand the following : (1×3)

(i) MTOC

(ii) GPCR

(iii) MAPK

2. Explain the cell cycle and discuss its regulation. Add a note on cell cycle checkpoints. (12)

3. Write an account of the components and functions of the mitochondrial respiratory chain. Add a note on the semi-autonomous nature of mitochondria. (12)

4. (a) Give an account of the packaging of chromosomal DNA in eukaryotic cells. (8)